Course Outline

Sem		Course Name	Th	Tut	Pra	Hrs	Cr
Sem I	HSC	Humanities – 1	2	0	0	2	2
	BSC	Mathematics – 1	3	1	0	4	4
	BSC	Physics	3	0	0	3	3
	BSC	Physics – Laboratory	0	0	2	2	1
	BSC	Environment Science	2	0	0	2	2
	ESC	Workshop Practices	0	0	2	2	1
	BSC	Biology for Engineers	2	0	0	2	2
	ESC	Computer Programming -I	1	0	0	1	1
	ESC	Computer Programming -I Laboratory	0	0	2	2	1
	HSC	Universal Human Values	1	0	0	1	1
	HSC	Indian Knowledge System	2	0	0	2	2
			16	**1**	**5**	**23**	**20**
Sem II	HSC	Humanities – 2	2	0	0	2	2
	BSC	Mathematics – 2	3	1	0	4	4
	BSC	Chemistry	3	0	0	3	3
	BSC	Chemistry – Laboratory	0	0	2	2	1
	ESC	Introduction to Electrical Engineering	3	0	0	3	3
	ESC	Introduction to Electrical Engineering-Laboratory	0	0	2	2	1
	ESC	Engineering Graphics	0	0	4	4	2
	HSC	Yoga, Health & Hygiene OR NCC/NSS	0	0	2	2	1
	HSC	Organizational Behaviour	1	0	0	1	1
	ESC	Computer Programming -II	1	0	0	1	1
	ESC	Computer Programming -II Laboratory	0	0	2	2	1
			13	**1**	**12**	**26**	**20**
Sem III	Pro	Civic and Social Service Internship	0	0	0	0	1
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	PC	Digital Circuits	3	0	0	3	3
	PC	Digital Circuits Laboratory	0	0	2	2	1
	PC	Data Structures and Algorithms	3	0	0	3	3
	PC	Data Structures and Algorithms Laboratory	0	0	2	2	1
	PC	Fundamentals of ICT	2	0	0	2	2
	PC	Electronics Devices and Circuits	2	0	0	2	2
	BSC	Discrete Mathematical Structures	3	1	0	4	4
			16	**1**	**4**	**21**	**20**
Sem IV	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 – Laboratory (Department Specific)	0	0	2	2	1
	OE	Open Elective 1 (From Other School)	3	0	0	3	3
	PC	Database Management Systems	3	0	0	3	3
	PC	Database Management Systems Laboratory	0	0	2	2	1
	PC	Principles of Programming Languages	3	0	0	3	3
	PC	Principles of Programming Languages Laboratory	0	0	2	2	1

	PC	Fundamentals of Signal Processing & Communication	3	1	0	4	4
	PC	Computer Organization & Microprocessor	3	0	0	3	3
	PC	Computer Organization & Microprocessor Laboratory	0	0	2	2	1
			17	**1**	**8**	**26**	**22**
Sem V	OE	Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3
	HSC	Engineering Economics	3	0	0	3	3
	PE	Program Elective 1	3	0	0	3	3
	PC	Theory of Computation & Compiler Design	3	0	0	3	3
	PC	Theory of Computation & Compiler Design Laboratory	0	0	2	2	1
	PC	Digital Signal Processing	3	0	0	3	3
	PC	Digital Signal Processing Laboratory	0	0	2	2	1
	PC	RF Engineering	3	0	0	3	3
	PC	RF Engineering Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VI	OE	Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 2	3	0	0	3	3
	PE	Program Elective 3	3	0	0	3	3
	PC	Operating Systems	3	0	0	3	3
	PC	Operating Systems Laboratory	0	0	2	2	1
	PC	Digital Communication	3	0	0	3	3
	PC	Digital Communication Laboratory	0	0	2	2	1
	PC	Computer Communication and Networking	3	0	0	3	3
	PC	Computer Communication and Networking Laboratory	0	0	2	2	1
			18	**0**	**6**	**24**	**21**
Sem VII	Pro	Summer Internship	0	0	0	0	2
	OE	Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE	Program Elective 4 (3-0-0)	3	0	0	3	3
	PE	Program Elective 5 (3-0-0)	3	0	0	3	3
	PC	Software Engineering Methodology	3	0	0	3	3
	PC	Software Engineering Methodology Laboratory	0	0	2	2	1
	PC	Embedded Systems	3	0	0	3	3
	PC	Embedded Systems Laboratory	0	0	2	2	1
	PC	Digital CMOS and VLSI Design	3	0	0	3	3
	PC	Digital CMOS and VLSI Design Laboratory	0	0	2	2	1
		Seminar					1
			18	0	6	24	24
Sem VIII		**Course Name**	**Th**	**Tut**	**Pra**	**Hrs**	**Cr**
	Pro	Major/Comprehensive Project					**12**

4th Semester

<Course Code>					< Database Management Systems >					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To learn the fundamentals of data models and to conceptualize and depict a database system using an ER diagram.
2. To make a study of SQL and relational database design.
3. To understand the internal storage structures which will help in physical DB design.
4. To know the fundamental concepts of transaction processing- concurrency control techniques and recovery procedure.

UNIT I: CONCEPTUAL MODELING INTRODUCTION	**12 Hrs.**
Introduction - General introduction to database systems; Database - DBMS distinction, approaches to building a database, data models, database management system, three-schema architecture of a database, challenges in building a DBMS, various components of a DBMS.E/R Model - Conceptual data modeling - motivation, entities, entity types, various types of attributes, relationships, relationship types, E/R diagram notation, examples. Relational Data Model - Concept of relations, schema-instance distinction, keys, referential integrity, and foreign keys. Converting the database specification in E/R notation to the relational schema.	
UNIT II: RELATIONAL APPROACH and SQL QUERY - BASICS	10 Hrs.
Relational algebra operators: selection, projection, cross product, various types of joins, division, example queries, tuple relation calculus, domain relational calculus, Introduction to SQL, Data Definition of SQL, Basic structure of SQL queries, Basic SQL operations (rename, string operations, order by, where clause), Set operations, Null values, Aggregate functions, Nested Subqueries, Modification of Database, JOIN expressions, Views, Integrity constraints, Data types and Schemas, Authorization.	
UNIT III: RELATIONAL DATABASE DESIGN	10 Hrs.
Dependencies and Normal forms, dependency theory - functional dependencies, Armstrong's axioms for FD's, closure of a set of FD's, minimal covers, definitions of 1NF, 2NF, 3NF and BCNF, decompositions and desirable properties of them, algorithms for 3NF and BCNF normalization, 4NF, and 5NF	
UNIT IV: TRANSACTION MANAGEMENT AND CONCURRENCY	10 Hrs.
Transaction processing: Transaction concept, transaction State, implementation of atomicity and durability, concurrent executions, serializability, recoverability. Concurrency Control: Lock-based protocols, timestamp-based protocols, validation-based protocols, multiple granularity, multiversion schemes, deadlock handling. Recovery: Failure classification, storage structure, recovery and atomicity, Log-Based recovery, shadow paging,recovery with concurrent transactions buffer management.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand the data models, schemas, instances, view levels and database architecture for voluminous data storage
CO2	:	Remember the concept of Relational Algebra and Relational Calculus from set theory to represent queries
CO3	:	Apply the concepts of SQL queries for data aggregation, calculations, views, sub-queries, embedded queries manipulation
CO4	:	Illustrate the definition of Functional Dependencies, Inference rules, and minimal sets of FD's to maintain data integrity.
CO5	:	Design the concepts of transaction, states, and ACID properties in data manipulation.
CO6	:	Design Build a database management system that satisfies relational theory and provides users with business queries.

TEXT/REFERENCE BOOKS

1. Silberschatz, Henry F. Korth, and S. Sudharshan, **"Database System Concepts"**, 7th Ed, Tata McGraw Hill.
2. J. Date, A. Kannan and S. Swamynathan, **"An Introduction to Database Systems"**, 8thed, Pearson Education
3. RamezElmasri and Shamkant B. Navathe, **"Fundamentals of Database Systems",** 7thEdition, Pearson/Addison-Wesley.

<Course Code>	< Database Management Systems Laboratory>
Teaching Scheme	Examination Scheme

L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. The SQL commands for data definition, manipulation, control and perform transactions in database systems.
2. The procedural language for implementation of functions, procedures, cursors and triggers using PL/SQL programs.
3. The logical design of a real time database system with the help of Entity Relationship diagrams.

LIST OF EXPERIMENTS	
1	Exercises on Installation and Accessing Database
2	Exercises on database definition language queries
3	Exercises on database Manipulation Language Queries.
4	Exercises on both DDL and DML Queries.
5	Exercises on tables with different types of constraints.
6	Exercises on Join Operations and Aggregate Functions in the Database.
7	Exercises on TCL (transaction control language) Queries to perform rollbacking and save point mechanisms.
8	Exercises on user accounts to grant privileges using Data Control language process.
9	Working with basic PL/SQL programming Examples
10	Working with PL/SQL Programming Functions

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Demonstrate database creation and manipulation concepts with the help of SQL queries.
CO2	:	Make use of inbuilt functions of SQL queries to perform data aggregations, subqueries, embedded queries and views
CO3	:	Apply key constraints on database for maintaining integrity and quality of data
CO4	:	Demonstrate normalization techniques by using referential key constraints.
CO5	:	Implement PL/SQL programs on procedures, cursors and triggers for enhancing the features of database system to handle exceptions.
CO6	:	Design database model with the help of Entity Relationship diagrams for a real time system or scenario.

TEXT/REFERENCE BOOKS

1. Silberschatz, Henry F. Korth, and S. Sudharshan, “Database System Concepts”, 7th Ed, Tata McGraw Hill.
2. J. Date, A. Kannan and S. Swamynathan, “An Introduction to Database Systems”, 8thed, Pearson Education
3. RamezElmasri and Shamkant B. Navathe, “Fundamentals of Database Systems”, 7thEdition, Pearson/Addison-Wesley.

<Course Code>					< Principles of Programming Languages >					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

4. To learn about various constructs and their respective comparisons in different high-level languages.
5. To understand the salient features in the landscape of programming languages.
6. To understand design concepts for programming languages
7. To study various programming paradigms.

UNIT I: INTRODUCTION TO LANGUAGE DESIGN CONCEPTS	**10 Hrs.**
Reasons for studying concepts of programming languages, programming domains, language evaluation criteria, influences on language design, language categories, language design trade-offs, implementation methods, programming environments, Programming language design, programming language processing. Data types: primitive data types, composite data types, recursive data types, implementation and storage representation of data types. Type binding, binding and binding times, type checking, type conversion, expressions, statements.	
UNIT II: OBJECT-ORIENTED PROGRAMMING CONCEPTS	**10 Hrs.**
Need of object-oriented paradigm, basic concepts of object-oriented programming (OOP), benefits of OOP. General characteristics for OOP, concepts - object, classes, messages, methods. Class Identification, object oriented as abstract data type. Data abstraction, encapsulation, polymorphism, inheritance, dynamic binding, abstract classes, interfaces, generic class, run time type identification. Dynamic Memory Management, Exception Handling	
UNIT III: PROCEDURAL PROGRAMMING	**10 Hrs.**
Introduction to procedures, parameter passing methods, lifetime of variables, scope rules: static and dynamic scope, nested scope, procedure call and return, recursive sub-program. Referencing environment, activation records, storage management, desirable and undesirable characteristics of procedural programming.	
UNIT IV: FUNCTIONAL PROGRAMMING	**10 Hrs.**
Introduction to functional programming, lambda calculus, ambiguity, free and bound identifiers, reductions, typed lambda calculus, application of functional programming. Functional Programming with python, elements of functional Programming, function declaration, expression evaluation, type checking.	
TOTAL HOURS: 40 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Identify fundamental concepts, terminologies, and syntax of various programming languages.
CO2	:	Understand the differences between various programming paradigms such as procedural, object-oriented, functional, and logical programming.
CO3	:	Apply the concepts programming languages to solve specific computational problems, applying appropriate syntax and constructs.
CO4	:	Analyze the impact of language design choices on program efficiency, maintainability, and scalability
CO5	:	Evaluate the suitability of different programming languages for specific application domains and development environments.
CO6	:	Design innovative solutions to complex problems using advanced language features and programming paradigms.

TEXT/REFERENCE BOOKS

1. S Roosta, "**Foundations of Programming Languages"**, Thomson, Brooke/Cole, (India Edition)
2. R Sethi, **"Programming Languages concepts & constructs",** Pearson Education, (2nd Edition)
3. R Sebesta, **"Concepts Of Programming Languages",** Pearson Education, (10th Edition)
4. Allen Tucker, Robert Noonan, "**Programming Languages: Principles and Paradigms",** Tata McGraw Hill, (2nd edition)

<Course Code>					< Principles of Programming Languages Laboratory>					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To introduce the various programming paradigms.
2. To understand the evolution of programming languages.
3. To understand the concepts of OO languages, functional languages, logical and scripting languages.

LIST OF EXPERIMENTS	
1	To study the concepts of programming languages.
2	To study the programs to implement the concepts of classes and objects.
3	To design a program to Demonstrate the concept of constructor.
4	To design a program to demonstrate Single, multiple, multilevel, hybrid, hierarchical inheritance, and Virtual base classes
5	To design a program to demonstrate the concept of operator overloading
6	To design a program to Demonstrate the concept of dynamic memory management
7	To design a program to demonstrate the concept of Exception Handling
8	To design a program to demonstrate the concept of Generic Programming
9	To design a program to demonstrate the concept of functional programming
10	To design a program to demonstrate the concept of procedural programming

COURSE OUTCOMES -

On completion of the course, student will be able to:

CO1	:	Introduce notations to describe syntax, semantics of programming languages and language features used in current programming languages.
CO2	:	Understand the major programming paradigms, and the principles and techniques involved in design and implementation of modern programming languages
CO3	:	Analyze the concepts of ADT, Object Oriented Programming
CO4	:	Analyze and explain behavior of simple programs in imperative languages using concepts such as binding, scope, control structures, subprograms and parameter passing mechanisms.
CO5	:	Evaluate key concepts in the implementation of common features of programming languages.
CO6	:	Demonstrate Concurrency Control and Exception Handling for large-scale software development.

TEXT/REFERENCE BOOKS

1. S Roosta, "**Foundations of Programming Languages**", Thomson, Brooke/Cole, (India Edition)
2. R Sethi, "**Programming Languages concepts & constructs**", Pearson Education, (2nd Edition)
3. R Sebesta, "**Concepts Of Programming Languages**", Pearson Education, (10th Edition)

<Course Code>					Fundamentals of Signal Processing and Communication					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

8. Introduce mathematical tools for analysis of continuous time signals and LTI systems.
9. Understand basic principles of communication systems.
10. Create foundation for digital signal processing.

UNIT I: CONTINUOUS TIME SIGNALS AND SYSTEMS	10 Hrs.
Signal – Definition, Examples, Classifications, Continuous time signals: Energy, power, periodicity, Signal operations, Elementary Signals, System - Definition, Classification, Continuous time LTI systems: Convolution integral and unit impulse response, Interconnections, stability and causality.	
UNIT II: FOURIER ANALYSIS AND CORRELATION OF SIGNALS	**13 Hrs.**
Trigonometric and Exponential Fourier series for Periodic Signals, Convergence and Truncation, Fourier transform, Signal spectra, Dirichlet conditions, Important Properties. Impulses in Frequency Domain and Fourier Transform of Periodic Signals, Frequency analysis of LTI systems, Signal distortions, Distortionless transmission, Ideal vs. practical filters, Parseval's relation, Essential bandwidth, Time correlation of signals, Spectral densities, Relation between convolution and time correlations.	
UNIT III: FUNDAMENTALS OF COMMUNICATION SYSTEMS	**11 Hrs.**
Baseband and carrier communication, AM Schemes: modulation and detection – DSBFC, DSBSC, SSB, VSB, QAM, Angle Modulation and Demodulation, Wideband and narrowband FM, Principles and working of super heterodyne receiver: sensitivity and selectivity, Noise types, Signal to noise ratio (SNR), Noise factor and Noise figure, Noise modeling, PSD of white noise and narrow band noise, Noise performance comparison of AM and FM schemes.	
UNIT IV: FOUNDATION FOR DIGITAL SIGNAL PROCESSING	**8 Hrs.**
Sampling and Sampling theorem, Reconstruction, Aliasing, Discrete time elementary signals, Discrete time LTI system and unit impulse (sample) response, Convolution sum, Linear constant co-efficient difference equation representation, Stability and causality, Discrete time vs. digital signal processing, Concept of quantization, A to D and D to A conversion, Future study directions.	
TOTAL HOURS: 42 Hrs.	

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Remember properties of signals, systems, and principles of various modulation and demodulation methods.
CO2	:	Understand mathematical representation of signals, systems, modulated signals and detection techniques.
CO3	:	Apply mathematical concepts to model behavior of LTI systems, modulation and demodulation of signals.
CO4	:	Analyze signals and systems in time and frequency domain to obtain various parameters.
CO5	:	Evaluate properties of LTI systems and performance of communication systems.
CO6	:	Design a block diagram level elementary communication and signal processing system.

TEXT/REFERENCE BOOKS

(1) Alan V.Oppenheim, Alan S.Willsky with S. Hamid Nawab, Signals & Systems, Pearson Education.
(2) B. P. Lathi, Signal Processing and Linear System, Berkeley Cambridge Press.
(3) B.P.Lathi, Zhi Ding, Modern Digital and Analog Communication Systems, Oxford University Press.
(4) P. Chakrabarti, Analog and Digital Communication, Dhanpat Rai & Co.
(5) J. G.Proakis, D. G.Manolakis, Digital Signal Processing, Principles, Algorithms and Applications, PHI.

24ICxxxT					Computer Organization and Microprocessor					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand design aspects of basic computer systems
- To explore the advanced features for throughput improvement
- To apply concepts in microprocessor system development
- To make basis for operating systems by understanding the concepts of memory management.

UNIT 1: Basic Computer Organization	**11Hrs.**
Introduction to FSM, Moore and Mealy machines. Basic architecture of micro-computer and Central processing unit, concept of data, address and control bus, register transfer language, Stored program organization, instruction format, common bus system, instruction set completeness, timing and control, instruction cycle, types of instructions, design of Basic central processing unit. Hardwired and microprogramed control Units, Interrupts in basic computer.	
UNIT 2: Advanced features of computer system	**10 Hrs.**
Flynn's classification of computer systems, Parallel processing, pipelining, vector processing, arithmetic and instruction pipeline. Memory organization, cache and virtual memory management.	
UNIT 3 : 8086 Microprocessor	**11 Hrs.**
Evolution of microprocessor, Pin diagram, signals, and register set of 8086. Addressing modes, instruction set, assembly language programming, Counters and Time delays, Stack, subroutine, Restart, Conditional Call and Return Instructions, interrupts of 8086.	
UNIT 4 : Interfacing using 8085	**10 Hrs.**
Address decoding techniques, interfacing concepts, IO mapped IO, Memory mapped IO, interfacing of basic devices, interfacing PPI 8255, and Interrupt controller 8259.	

Total: 42Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Recall and describe the basic components of computer architecture, including the CPU, memory, and I/O devices, and understand their respective functions and interactions within a computer system.

CO2: Demonstrate the understanding of basic computer design, architecture and functionalities of the 8085 microprocessor, including its instruction set, addressing modes, and interfacing with peripherals.

CO3: Practice and debug assembly language programs for the 8085 microprocessor to perform specific tasks, demonstrating their ability to apply the knowledge of microprocessor architecture in practical programming scenarios.

CO4: Apprise the concept of programming, including data types, control structures, functions, and algorithms, and will be able to analyze and explain the functionality of different programming constructs

CO5: Evaluate the advanced architectural features and innovations in modern CPUs, such as pipelining, caching, multi-core processing, and parallel computing, and understand their impact on performance and efficiency.

CO6: Integrate and apply their knowledge of basic computer architecture design, 8085 microprocessor architecture, programming, and advanced CPU features to design, develop, and optimize a basic computer system or program, demonstrating their ability to create and innovate in the field of computer architecture and design.

TEXT/REFERENCE BOOKS

1. Morris M. Mano, "Computer System Architecture", Pearson Education, 3rd edition, 2017.
2. John P. Hayes, "Computer Architecture and Organization", McGraw Hill Education, 3rd edition, 1998.
3. William Stallings, "Computer Organization & Architecture", Pearson Education, 11th Edition, 2022.
4. Ramesh S. Gaonkar, "Microprocessor Architecture, Programming, and Applications with 8085", Penran International, 6th edition, 2013.
5. N. Senthil Kumar, M. Saravanan, S. Jeevanathan, S. K. Shah, "Microprocessors and Interfacing", Oxford University Press, 2012.
6. Douglas Hall, "Microprocessor and Interfacing", Mcgraw Hill Higher Education, 2005.

24ICTxxxP	Computer Organization and Microprocessor Lab

Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	0	0	0	50	50	100

COURSE OBJECTIVES:

- To understand design aspects of basic computer systems through simulation
- To apply concepts in microprocessor system development
- To do assembly language programming using 8085 instruction set to complete different tasks.

List of Experiments: Following is the list of experiments but it is not limited to:

1. To design a FSM with given sequence
2. To design and demonstrate the working of a massage on 7-segmnet display.
3. To design register with synchronous control inputs like clear, load and increment and with provision to handle different no of bit than common bus.
4. To design common bus systems using multiplexure
5. To design ALU with given specification
6. To design control unit for given CPU specification
7. To Write an assembly language code to implement data transfer instruction.
8. To write an assembly language code to store numbers in reverse order in memory location.
9. To write an assembly language code to implement arithmetic instruction.
10. To write an assembly language code to implement logical instruction set to carry out a task.
11. To write an assembly language code to implement the subroutine call.
12. To write an assembly language code to implement RIM and SIM to interface a peripheral.

*** Some of these experiments need to be using simulation software.**

COURSE OUTCOMES:

On completion of the course, student will be able to

CO1: Recall and describe the basic components of computer architecture, including the CPU, memory, and I/O devices, and understand their respective functions and interactions within a computer system.

CO2: Demonstrate the understanding of basic computer design, architecture and functionalities of the 8085 microprocessor, including its instruction set, addressing modes, and interfacing with peripherals.

CO3: Practice and debug assembly language programs for the 8085 microprocessor to perform specific tasks, demonstrating their ability to apply the knowledge of microprocessor architecture in practical programming scenarios.

CO4: Apprise the concept of programming, including data types, control structures, functions, and algorithms, and will be able to analyze and explain the functionality of different programming constructs

CO5: Evaluate the advanced architectural features and innovations in modern CPUs, such as pipelining, caching, multi-core processing, and parallel computing, and understand their impact on performance and efficiency.

CO6: Integrate and apply their knowledge of basic computer architecture design, 8085 microprocessor architecture, programming, and advanced CPU features to design, develop, and optimize a basic computer system or program, demonstrating their ability to create and innovate in the field of computer architecture and design.

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1. Morris M. Mano, “Computer System Architecture”, Pearson Education, 3rd edition, 2017.
2. John P. Hayes, “Computer Architecture and Organization”, McGraw Hill Education, 3rd edition, 1998.
3. William Stallings, “Computer Organization & Architecture”, Pearson Education, 11th Edition, 2022.
4. Ramesh S. Gaonkar, “Microprocessor Architecture, Programming, and Applications with 8085”, Penran International, 6th edition, 2013.
5. N. Senthil Kumar, M. Saravanan, S. Jeevanathan, S. K. Shah, “Microprocessors and Interfacing”, Oxford University Press, 2012.
6. Douglas Hall, “Microprocessor and Interfacing”, Mcgraw Hill Higher Education, 2005.